Gross Anatomy of the Oral Cavity Set 1)

1. The oral cavity is anatomically split into two distinct functional zones. The outer slit-like space situated between the lips or cheeks externally and the teeth or gingivae internally is the:
a. Oral cavity proper
b. Fauces
c. Oropharyngeal isthmus
d. Oral vestibule
e. Sublingual space
2. The oral vestibule communicates with the exterior of the face via the oral fissure. When a patient clenches their teeth tightly, the vestibule still communicates with the oral cavity proper via a space located behind which teeth?
a. Central incisors
b. Maxillary canines
c. Mandibular premolars
d. Maxillary first premolars
e. The last molar teeth (retromolar space)
3. The upper and lower lips are mobile fibroelastic folds. Near the midline, the internal mucosal surface of each lip is connected to the adjacent gingiva by a specialized fold of mucous membrane called the:
a. Pterygomandibular raphe
b. Labial frenulum
c. Lingual frenulum
d. Alveolar mucosa
e. Sublingual fold
4. The cheeks form the lateral walls of the oral vestibule. The primary muscle that forms the fleshy substance of the cheek, compressing it against the teeth to prevent food stagnation during chewing, is the:
a. Masseter muscle
b. Buccinator muscle
c. Medial pterygoid muscle
d. Temporalis muscle
e. Orbicularis oris
5. The buccinator muscle is wrapped externally by a distinct layer of fascia that continues posteriorly over the superior pharyngeal constrictor. This layer is the:
a. Buccopharyngeal fascia
b. Investing cervical fascia
c. Pretracheal fascia
d. Deep temporal fascia
e. Masseteric fascia
6. The parotid salivary duct (Stensen's duct) passes forward across the masseter muscle and turns medially to pierce the buccinator muscle. It opens into the oral vestibule opposite the crown of the:
a. Maxillary first premolar
b. Maxillary second molar
c. Mandibular first molar

d. Maxillary third molar
e. Mandibular second molar

7. When the parotid duct pierces the buccinator muscle and oral mucosa, its clinical opening is often marked by a small soft tissue elevation known as the:

a. Sublingual caruncle
b. Plica fimbriata
c. Parotid papilla
d. Pterygomandibular fold
e. Retromolar pad

8. The tissue of the lips displays a transitional zone between the external skin and the internal mucous membrane. This hairless, reddish zone is known as the:

a. Philtrum
b. Labial commissure
c. Mucobuccal fold
d. Vermilion border (zone)
e. Nasolabial sulcus

9. The shallow vertical groove seen in the midline of the upper lip on the external skin surface, running from the nasal septum down to the vermilion border, is the:

a. Labial frenulum
b. Mentolabial sulcus
c. Philtrum
d. Nasolabial groove
e. Incisive fossa

10. The dense submucosa of the upper and lower lips contains a continuous ring of small, round, mucus-secreting minor salivary glands that can be easily palpated. These are the:

a. Buccal glands
b. Palatine glands
c. Labial glands
d. Von Ebner glands
e. Molar glands

11. The gingiva (gum) is divided into multiple topographical zones. The tight collar of gingiva that surrounds the neck of the tooth but is not directly attached to the bone surface is the:

a. Attached gingiva
b. Free (marginal) gingiva
c. Interdental papilla
d. Alveolar mucosa
e. Mucogingival junction

12. The specialized segment of the gingiva that fills the interproximal space between two adjacent contacting teeth is formally termed the:

a. Attached gingiva
b. Free gingival groove
c. Mucobuccal fold
d. Interdental papilla
e. Junctional epithelium

13. Extending apically from the free gingival groove to the mucogingival junction is a firm, immobile zone of masticatory mucosa that is tightly bound to the alveolar bone periosteum, named the:
a. Alveolar mucosa
b. Vestibular sulcus
c. Attached gingiva
d. Marginal gingiva
e. Interdental col
14. Healthy attached gingiva often exhibits a characteristic textured, dimpled, or orange-peel appearance macroscopically. This clinical feature is known as:
a. Striation
b. Keratin tufting
c. Stippling
d. Trabeculation
e. Papillomatosis
15. At the apical boundary of the attached gingiva, there is a sharp structural line where the firm pink masticatory mucosa transitions into loose, dark red, highly vascular tissue. This line is the:
a. Free gingival groove
b. Mucogingival junction
c. Gingival sulcus
d. Alveolar crest line
e. Labial commissure
16. The loose, movable, non-keratinized mucosal lining that covers the basal part of the alveolar process and continues into the vestibular sulcus is the:
a. Attached gingiva
b. Marginal gingiva
c. Alveolar mucosa
d. Interdental papilla
e. Junctional band
17. The primary arterial blood supply to the facial (buccal) gingiva of the mandibular molar teeth is derived from branches of the:
a. Greater palatine artery
b. Lingual artery
c. Buccal artery (branch of the maxillary artery)
d. Superior labial artery
e. Incisive artery
18. The deep, narrow space or gutter bounded by the tooth surface on one side and the internal lining of the free gingiva on the other side is the:
a. Alveolar socket
b. Gingival sulcus
c. Interdental col
d. Mucobuccal fold
e. Retromolar fossa
19. In a clinically healthy mouth, the normal biological depth of the gingival sulcus when measured with a periodontal probe varies between:
a. 1 to 3 mm
b. 5 to 7 mm

c. 8 to 10 mm
d. 0.1 mm exactly
e. 12 to 15 mm

20. The floor of the oral cavity proper is formed primarily by a broad, sheet-like muscular partition that stretches between the mylohyoid lines of the mandible and the hyoid bone. This partition is the:

a. Geniohyoid muscle
b. Mylohyoid muscle
c. Genioglossus muscle
d. Hyoglossus muscle
e. Digastric muscle anterior belly

21. The mylohyoid muscle is functionally known as the diaphragm of the mouth. It is supplied motorically by the mylohyoid nerve, which arises from the:

a. Lingual nerve
b. Inferior alveolar nerve (branch of V3)
c. Facial nerve cervical branch
d. Hypoglossal nerve
e. Ansa cervicalis

22. Located on either side of the midline in the anterior floor of the mouth is a small, elevated mucosal projection that marks the opening of the submandibular salivary duct. This projection is the:

a. Sublingual fold (plica sublingualis)
b. Sublingual caruncle (papilla)
c. Retromolar papilla
d. Parotid papilla
e. Mental tubercle

23. The submandibular salivary duct (Wharton's duct) emerges from the deep part of the submandibular gland and runs forward along the floor of the mouth. During its course, it is crossed loops-and-turns style by the:

a. Hypoglossal nerve
b. Mylohyoid nerve
c. Lingual nerve
d. Facial artery
e. Glossopharyngeal nerve

24. Running transversely across the floor of the mouth, lateral to the sublingual caruncle, is a ridge produced by the underlying sublingual salivary gland and its multiple small ducts. This is the:

a. Sublingual caruncle
b. Plica fimbriata
c. Sublingual fold (plica sublingualis)
d. Sulcus terminalis
e. Retromolar triangle line

25. The submandibular salivary gland is located partially inside the submandibular triangle. Its primary duct opens into the floor of the mouth, but a portion of the gland hooks around the posterior border of which muscle?

a. Masseter muscle
b. Buccinator muscle
c. Mylohyoid muscle

d. Geniohyoid muscle
e. Stylohyoid muscle

26. The sublingual salivary gland sits immediately beneath the oral mucosa of the floor of the mouth. It drains its secretions into the oral cavity via several small, short ducts known as the:

a. Ducts of Rivinus (and the major duct of Bartholin)
b. Ducts of Stensen
c. Ducts of Wharton
d. Ducts of Gasser
e. Ducts of Von Ebner

27. Human dentition is heterodont, meaning it contains different classes of teeth. The permanent dental quadrant formula for one-half of the human adult mouth displays which sequence of teeth?

a. 2 Incisors, 1 Canine, 2 Molars
b. 2 Incisors, 1 Canine, 2 Premolars, 3 Molars
c. 3 Incisors, 1 Canine, 1 Premolar, 2 Molars
d. 2 Incisors, 2 Canines, 1 Premolar, 3 Molars
e. 1 Incisor, 1 Canine, 2 Premolars, 4 Molars

28. Primary (deciduous) dentition is different from permanent dentition. The primary dentition completely lacks which entire class of teeth?

a. Incisors
b. Canines
c. Premolars
d. Molars
e. Wisdom teeth

29. The total number of teeth present in a complete, normal deciduous dentition is:

a. 16 teeth
b. 20 teeth
c. 28 teeth
d. 32 teeth
e. 10 teeth

30. The total number of teeth present in a complete, normal permanent adult dentition is:

a. 20 teeth
b. 24 teeth
c. 32 teeth
d. 40 teeth
e. 16 teeth

31. Permanent maxillary molars have a unique root morphology. Under normal anatomical conditions, they typically possess how many distinct roots?

a. One single fused root
b. Two roots (mesial and distal)
c. Three roots (two buccal, one palatal)
d. Four roots matching four cusps
e. Five accessory roots

32. Permanent mandibular molars differ structurally from permanent maxillary molars because they typically possess only:

a. One single fused root

b. Two distinct roots (mesial and distal)
c. Three distinct roots
d. Four small roots
e. A massive palatal root only

33. The permanent tooth that possesses the longest, thickest root in the entire dental arch, producing a prominent skeletal ridge on the face of the maxilla, is the:

a. Maxillary central incisor
b. Maxillary first molar
c. Maxillary canine
d. Mandibular first premolar
e. Maxillary second premolar

34. The specialized bony sockets within the alveolar processes of the maxilla and mandible that accommodate the roots of the teeth are known anatomically as the:

a. Dental laminae
b. Alveoli (dental sockets)
c. Odontogenic crypts
d. Trabecular meshes
e. Mental foramina

35. The hard fibrous connective tissue structure that anchors the cementum of the tooth root directly to the surrounding alveolar bone proper, acting as a shock absorber, is the:

a. Gingival fiber group
b. Periodontal ligament (PDL)
c. Sphenomandibular ligament
d. Dental pulp core
e. Periapical cementum matrix

36. The mineralized tissue that forms the bulk or core of the tooth structure, running through both the crown and the root, is:

a. Enamel
b. Cementum
c. Dentin
d. Pulp
e. Alveolar bone

37. The crystalline, highly calcified tissue that covers the outer surface of the anatomical crown of a tooth is:

a. Dentin
b. Enamel
c. Cementum
d. Cortical bone
e. Osteoid tissue

38. The thin, specialized calcified tissue that covers the outer surface of the anatomical root of a tooth is:

a. Enamel
b. Dentin
c. Cementum
d. Periodontal bone
e. Alveolar cortex

39. The central, soft tissue compartment of a tooth that contains the blood vessels, lymphatics, and nerves is the:
a. Dental pulp (housed in the pulp chamber and root canals)
b. Periodontal space
c. Gingival sulcus
d. Alveolar trabeculae
e. Dentinal tubule canal
40. Sensory innervation to all permanent maxillary teeth is mediated by branches of the maxillary nerve (V2). Specifically, the maxillary molar teeth are supplied by the:
a. Anterior superior alveolar nerve
b. Middle superior alveolar nerve
c. Posterior superior alveolar nerve
d. Infraorbital nerve terminal loops
e. Nasopalatine nerve
41. All permanent mandibular teeth receive their sensory nerve supply from a single major branch of the mandibular nerve (V3). This branch is the:
a. Lingual nerve
b. Inferior alveolar nerve
c. Long buccal nerve
d. Mylohyoid nerve
e. Mental nerve
42. The inferior alveolar nerve enters the mandibular ramus via the mandibular foramen. Before entering the bone, it runs inside a fascial space bounded laterally by the ramus and medially by the:
a. Masseter muscle
b. Medial pterygoid muscle
c. Lateral pterygoid muscle
d. Temporalis muscle
e. Superior pharyngeal constrictor
43. To perform an inferior alveolar nerve block for lower tooth extractions, the dental surgeon inserts the needle lateral to a vital soft tissue band that stretches from the pterygoid hamulus to the mylohyoid line. This band is the:
a. Stylohyoid ligament
b. Sphenomandibular ligament
c. Pterygomandibular raphe
d. Free gingival groove
e. Sublingual fold
44. If the dental needle is accidentally directed too far posteriorly during an inferior alveolar nerve block, the anesthetic solution will enter the parotid gland capsule and cause:
a. Total numbness of the tongue
b. Severe pain in the temporal fossa
c. Transient facial paralysis (inability to close the eye or smile on that side)
d. Profuse salivation from the parotid duct
e. Spasm of the masseter muscle (trismus)
45. Following the extraction of a permanent tooth, the lack of mechanical stress causes a specific physiological alteration in the bone structure of the alveolar process. This process is:

a. Hypertrophy of the cortical plate
b. Atrophy and bone resorption of the alveolar ridge
c. Elongation of the condyle
d. Ossification of the mandibular canal
e. Hyperplasia of the mental spines

46. A patient experiences a severe spasm of the muscles of mastication, making it impossible to open the mouth. This clinical dental condition is known as:
a. Epiphora
b. Trismus (lockjaw)
c. Miosis
d. Ptosis
e. Ophthalmoplegia

47. The small triangular area of bone located immediately posterior to the lower permanent third molar tooth, bounded by the buccal and temporal crests, is the:
a. Sublingual fossa
b. Retromolar fossa (triangle / pad)
c. Pterygoid fovea
d. Mandibular notch
e. Digastric basin

48. The floor of the maxillary sinus sits in close anatomical proximity to the roots of the maxillary molars. A root fragment can accidentally be pushed upward during an extraction, creating an:
a. Oronasal fistula
b. Oroantral communication (or fistula)
c. Tracheoesophageal tract
d. Branchial cleft fistula
e. Sublingual abscess track

49. The severe pain of a toothache from a mandibular molar can sometimes be referred to the ear and temple region because the lower teeth and the auriculotemporal nerve share origins from the:
a. Ophthalmic nerve (V1)
b. Maxillary nerve (V2)
c. Mandibular nerve (V3)
d. Facial nerve (CN VII)
e. Glossopharyngeal nerve (CN IX)

50. The structural muscle that attaches to the pterygomandibular raphe anteriorly and blends into the lateral walls of the oral vestibule is the:
a. Superior pharyngeal constrictor
b. Masseter muscle
c. Buccinator muscle
d. Mylohyoid muscle
e. Orbicularis oris

Answer Key Table (Revised Oral Cavity Gross Anatomy - Set 1)

Q 1-8	Q 9-16	Q 17-24	Q 25-32	Q 33-40	Q 41-48	Q 49-50
d	c	c	c	c	b	c
e	c	b	c	b	c	c
b	b	b	a	e	b	
b	c	b	b	b	c	
a	b	c	c	b	b	
b	c	c	b	c	b	
c	b	c	d	a	b	
d	c	d	b	c	b	